

Table 3 — Sensitivity: Blade Discretisation (TSR=8)

N	dist.	CT	CP	■ [-]	dCT	dCP	iters	conv.
5	Cosine	0.6680	0.4692	0.2322	+0.0089	+0.0148	23	✓
5	Constant	0.6753	0.4791	0.2292	+0.0162	+0.0247	19	✓
8	Cosine	0.6661	0.4639	0.2432	+0.0070	+0.0095	21	✓
8	Constant	0.6679	0.4695	0.2310	+0.0088	+0.0151	23	✓
10	Cosine	0.6626	0.4597	0.2436	+0.0035	+0.0054	28	✓
10	Constant	0.6666	0.4669	0.2335	+0.0075	+0.0125	22	✓
13	Cosine	0.6620	0.4584	0.2469	+0.0029	+0.0040	59	✓
13	Constant	0.6658	0.4647	0.2372	+0.0066	+0.0103	23	✓
15	Cosine	0.6605	0.4564	0.2480	+0.0014	+0.0020	47	✓
15	Constant	0.6656	0.4638	0.2396	+0.0065	+0.0094	27	✓
18	Cosine	0.6597	0.4553	0.2490	+0.0006	+0.0009	32	✓
18	Constant	0.6652	0.4624	0.2430	+0.0060	+0.0080	26	✓
20	Cosine	0.6591	0.4544	0.2503	+0.0000	+0.0000	49	✓
20	Constant	0.6640	0.4607	0.2436	+0.0049	+0.0063	33	✓
25	Cosine	0.6585	0.4536	0.2514	-0.0006	-0.0008	122	✓
25	Constant	0.6623	0.4584	0.2450	+0.0032	+0.0040	21	✓
30	Cosine	0.6577	0.4525	0.2524	-0.0014	-0.0019	99	✓
30	Constant	0.6616	0.4572	0.2465	+0.0025	+0.0028	23	✓

N	dist.	CT	CP	■ [-]	dCT	dCP	iters	conv.
35	Cosine	0.6572	0.4518	0.2531	-0.0020	-0.0026	137	✓
35	Constant	0.6608	0.4559	0.2480	+0.0017	+0.0015	60	✓
40	Cosine	0.6567	0.4511	0.2536	-0.0024	-0.0033	473	✓
40	Constant	0.6603	0.4552	0.2489	+0.0012	+0.0008	27	✓
50	Cosine	0.6558	0.4503	0.2541	-0.0033	-0.0041	231	✓
50	Constant	0.6593	0.4537	0.2503	+0.0002	-0.0006	421	✓
60	Cosine	0.6548	0.4491	0.2539	-0.0043	-0.0053	612	✓
60	Constant	0.6582	0.4526	0.2509	-0.0009	-0.0018	26	✓
70	Cosine	0.6547	0.4489	0.2545	-0.0044	-0.0055	894	✓
70	Constant	0.6573	0.4515	0.2511	-0.0018	-0.0029	135	✓
80	Cosine	0.6540	0.4482	0.2542	-0.0051	-0.0062	1000	✗
80	Constant	0.6578	0.4516	0.2529	-0.0013	-0.0028	508	✓
90	Cosine	0.6538	0.4480	0.2544	-0.0053	-0.0064	1000	✗
90	Constant	0.6565	0.4505	0.2521	-0.0026	-0.0039	51	✓
100	Cosine	0.6534	0.4475	0.2546	-0.0057	-0.0069	1000	✗
100	Constant	0.6564	0.4502	0.2528	-0.0027	-0.0042	69	✓
125	Cosine	0.6528	0.4461	0.2543	-0.0063	-0.0083	1000	✗
125	Constant	0.6550	0.4487	0.2530	-0.0041	-0.0056	110	✓
150	Cosine	0.6527	0.4470	0.2548	-0.0064	-0.0074	1000	✗
150	Constant	0.6547	0.4481	0.2535	-0.0044	-0.0063	253	✓

Reference: $N=20$, cosine. Other params: $N_{\text{wake}}=5$, $d\psi=10.0$ deg, a_w mode=fixed=0.25. $\bar{a}$ = span-averaged axial induction.